

# FurFeel - Classification Models

## 1. Posture & Motion Activity (Random Forest)

The model determining the physical posture of the dog (e.g., walking, resting, shaking) is a Random Forest Classifier.

- Runs directly on the Arduino firmware (Collar) in real-time.
- Uses 20Hz IMU data (Accelerometer and Gyroscope).
- Data is chunked into 1-second windows.
- Features extracted per window: mean, standard deviation, min, max, and magnitude across 6 axes.

## 2. Stress Classification (Rule-Based Engine)

Stress classification is calculated in the cloud via a deterministic Rule-Based Engine (rule-v1), NOT machine learning. This provides maximum explainability so owners know EXACTLY why their dog is stressed.

The algorithm uses a point-based scoring system based on real-time telemetry versus the dog's personalized baseline.

### Scoring Rules Table

| Rule                        | Condition                          | Points |
|-----------------------------|------------------------------------|--------|
| Heart Rate (Elevated)       | 1.15x to 1.35x over baseline       | +1     |
| Heart Rate (Elevated)       | 1.35x to 1.6x over baseline        | +2     |
| Heart Rate (Elevated)       | Over 1.6x over baseline            | +3     |
| Respiratory Rate (Elevated) | 1.3x to 1.8x over baseline         | +1     |
| Respiratory Rate (Elevated) | Over 1.8x over baseline            | +2     |
| Motion (Restlessness)       | Motion score 0.6 to 0.8            | +1     |
| Motion (Restlessness)       | Motion score > 0.8                 | +2     |
| Posture Combo               | Moving + High Motion (>0.6)        | +1     |
| Environmental Amp.          | Temp > 32C OR Humidity > 80%       | +1     |
| Rising Trend                | Rising over 3 consecutive readings | +1     |

### Final Stress Level Thresholds

| Level    | Score Range  |
|----------|--------------|
| Calm     | 0 - 1 points |
| Mild     | 2 - 3 points |
| Moderate | 4 - 6 points |
| High     | 7+ points    |